

RogueGPT: A Controlled Stimulus Generation Framework for News Authenticity Research

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Summary

RogueGPT is an open-source Python framework for the controlled, reproducible generation and curation of multilingual news fragments for AI authenticity research. It enables researchers to systematically produce synthetic news stimuli across a wide range of large language model (LLM) families, journalistic styles, languages, and content formats, while storing every fragment alongside complete generation provenance in a MongoDB corpus.

The framework follows a three-layer architecture that strictly separates the data logic (`core.py`) from user-facing interfaces: a Streamlit web application, a command-line interface (CLI), and a Model Context Protocol (MCP) server for AI-agent integration. All three interfaces share a single validation and normalisation layer, ensuring that every fragment — whether ingested manually, via automated generation scripts, or through an AI agent — conforms to the same schema and that all experimental parameters are recorded for reproducibility.

The current corpus contains 3,278 multilingual news fragments. Of these, 2,638 are machine-generated by 37 model configurations across 10 providers (OpenAI, Anthropic, Google, Meta, Mistral, DeepSeek, Microsoft, Zhipu, Moonshot, and Qwen), covering four languages (English, German, French, and Spanish), three content formats (tweet, headline, and short article), and five journalistic styles per language. The remaining 640 fragments are human-sourced — both legitimate news and authentic fake-news material — and serve as experimental anchors for perception studies.

RogueGPT is the upstream stimulus generation component of a three-tool research pipeline: CRED-1 ([Loth et al., 2025](#)) identifies unreliable news sources, RogueGPT generates controlled stimuli, and JudgeGPT delivers them to human participants for perception measurement.

Statement of Need

Empirical research on human and automated detection of AI-generated news requires stimuli that are diverse, systematically varied across experimental conditions, and fully reproducible. Existing corpora are typically static snapshots generated by a single model or a narrow set of models, lacking the fine-grained metadata required to isolate the effect of individual experimental variables (model family, version, prompt strategy, language, style) on detection outcomes ([Kreps et al., 2022](#); [Zellers et al., 2019](#)).

RogueGPT addresses this gap in three ways:

- Systematic parameterisation.** Stimuli are generated by Cartesian expansion of a declarative configuration (`prompt_engine.json`) that defines models, languages, styles, and formats. Every combination is reachable without manual intervention, enabling researchers to expand the corpus to new model releases in minutes by updating a single JSON file.

2. **Complete provenance.** Every fragment is stored with the exact model identifier, prompt text, ISO language code, content format, journalistic style, ingestion channel, and a UTC creation timestamp. This makes it possible to reproduce any stimulus exactly and to filter the corpus along any experimental axis at query time via the CLI or MCP API.

3. **Living corpus design.** Unlike frozen datasets, the corpus is designed to grow. The MCP server exposes `ingest_fragment` and `retrieve_fragments` as tool-use endpoints, enabling AI agents to autonomously extend the corpus with new model outputs as part of larger automated pipelines. This is particularly important given the rapid pace of LLM releases: a framework that supports incremental corpus expansion without re-running entire generation pipelines is essential for longitudinal research.

The framework is part of a broader open-source research ecosystem for studying online misinformation. Upstream, the CRED-1 Domain Credibility Dataset (Loth et al., 2025) provides the automated source selection that identifies unreliable domains whose content is scraped as human-authored fake news anchors. Downstream, JudgeGPT delivers RogueGPT-generated stimuli to human participants and records dual-axis perception judgements. Together, the three tools form an end-to-end pipeline from source identification through stimulus generation to quantitative perception measurement.

RogueGPT is already in active use in peer-reviewed publications from the same research programme:

- *Industrialized Deception* (Loth, Kappes, et al., 2026d) analyses the systemic effects of LLM-generated misinformation on digital ecosystems, using stimuli generated by RogueGPT (ACM WWW '26).
- *Eroding the Truth-Default* (Loth, Kappes, et al., 2026a) reports human perception data from 320 participants who evaluated stimuli produced by this framework (ACM WWW '26).
- *The Verification Crisis* (Loth, Kappes, et al., 2026b) presents expert survey findings on GenAI disinformation and reproducible provenance as part of the same research programme (ACM WWW '26).
- The foundational survey *Blessing or Curse?* (Loth et al., 2024) describes the methodology underpinning the generation pipeline.

The full corpus produced by RogueGPT is archived on Zenodo (DOI: [10.5281/zenodo.18703138](https://doi.org/10.5281/zenodo.18703138)) and available for academic research under restricted access.

Key Features

Multi-Model Support

The model registry in `prompt_engine.json` currently defines 37 identifiers across 10 providers. The naming convention (`provider_model-variant`) is provider-agnostic; adding a new model requires a single line change to the JSON file and no code modifications. At ingest time, the CLI and MCP server validate submitted model identifiers against the registry, with an optional `--lenient` flag for provisional use of unreleased models.

Structured Fragment Schema

Each fragment is stored with complete provenance metadata including model identifier, prompt text, ISO language code, content format, journalistic style, source outlet, and UTC timestamp. A shared validation layer enforces schema consistency across all three interfaces, ensuring that every fragment — regardless of ingestion path — carries the metadata required to reproduce experimental conditions exactly. The full schema specification is documented in the repository.

85 Interfaces

86 RogueGPT provides three user-facing interfaces that share a single validation layer: a command-
87 line interface for scripted ingestion and retrieval, a Streamlit web application for interactive
88 use, and a Model Context Protocol (MCP) server that enables LLM-based agents to query the
89 corpus and ingest new fragments programmatically. The MCP integration allows AI agents to
90 autonomously extend the corpus as part of larger automated research pipelines.

91 Human Fragment Sourcing

92 Human-written fragments are sourced from two pools:

- 93 ■ **Legitimate news:** Opening paragraphs from established international outlets (BBC,
94 Guardian, Reuters, Tagesschau, FAZ, Le Monde, El País) via RSS feeds.
- 95 ■ **Fake news:** Articles scraped from domains classified as *fake*, *unreliable*, or *conspiracy* by
96 the CRED-1 Domain Credibility Dataset (Loth et al., 2025), which scores 2,672 domains
97 using editorial classification, fact-check claims, web popularity, and domain age.

98 Both categories are ingested with full provenance metadata, providing experimental anchors
99 that reflect authentic disinformation language rather than artificially constructed examples.

100 Software Design

101 RogueGPT is organised in three layers so that the data logic can be tested and reused
102 independently of any interface.

103 `core.py` is the only module that talks to MongoDB. It owns schema validation, field
104 normalisation, config loading, and the retrieval API. It has no UI dependencies, which is what
105 makes the test suite able to cover it directly.

106 Three interfaces sit on top of `core.py` and share its validation path: a Streamlit web application
107 (`app.py`) for interactive generation, a command-line interface (`cli.py`) for scripted and batch
108 work, and a Model Context Protocol server (`mcp_server.py`) that exposes `ingest_fragment`
109 and `retrieve_fragments` as tools for AI-agent workflows. Because all three route through the
110 same validator, a fragment ingested by an agent is subject to exactly the same schema and
111 provenance requirements as one entered by hand.

112 Generation behaviour is not hard-coded. `prompt_engine.json` declares the model identifiers,
113 prompt templates, languages, journalistic styles, and content formats, so extending the corpus
114 to a newly released model is a configuration change rather than a code change. Every stored
115 fragment carries the parameters that produced it, which is what allows a later analysis to filter
116 by any single experimental variable.

117 State of the Field

118 Several tools and datasets address fake news generation and detection research, but none
119 provide an active, extensible generation infrastructure with full provenance tracking.

120 GROVER (Zellers et al., 2019) generates and detects neural fake news but is limited to a
121 single GPT-2-scale architecture and English text. The LIAR dataset (W. Y. Wang, 2017)
122 provides 12,836 human-labelled claims across six fine-grained labels but offers no generation
123 infrastructure. TruthfulQA (Lin et al., 2022) measures model truthfulness across 817 questions
124 but targets factual consistency rather than controlled stimulus generation. Kreps et al. (Kreps
125 et al., 2022) demonstrate that GPT-2-generated political news is perceived as credible as
126 human-written content, establishing the need for systematic multi-model comparisons — but
127 their work uses a single model without a reusable generation framework.

128 More recent efforts such as the M4 corpus (Y. Wang et al., 2024) collect multi-generator
129 outputs across domains and languages but distribute them as static datasets without tooling
130 for reproducible expansion. Similarly, MULTITuDE (Macko et al., 2023) covers 11 generators
131 across 11 languages but is frozen at its release configuration. FakeNewsNet (Shu et al., 2020)
132 provides a rich repository combining news content with social context but focuses on detection
133 research with pre-collected articles rather than controlled stimulus generation.

134 RogueGPT differs from all of these in three respects: (1) it is an active generation infrastructure
135 rather than a frozen benchmark, supporting incremental corpus expansion as new models are
136 released; (2) it stores complete generation provenance per fragment, enabling fine-grained
137 experimental filtering by model, language, style, and format; and (3) its MCP integration is,
138 to our knowledge, the first such interface for a research corpus management system, enabling
139 direct use within AI agent workflows.

140 Research Impact Statement

141 RogueGPT exists to measure how people and detectors respond to AI-generated news, and its
142 design follows from that purpose. The framework produces experimental stimuli for controlled
143 studies; it has no publishing, posting, or distribution path, and no scheduling, targeting, or
144 audience-selection functionality. Fragments are written to a private MongoDB corpus and are
145 read back by researchers.

146 The scientific value lies in control and provenance. Because every fragment records the model,
147 prompt strategy, language, style, and format that produced it, effects can be attributed to a
148 specific variable rather than to an undifferentiated notion of “AI text”. This is what static,
149 single-model corpora cannot support, and it is the property that downstream perception work
150 depends on. RogueGPT is the upstream stage of a three-tool pipeline: CRED-1 (Loth et al.,
151 2025) identifies unreliable sources, RogueGPT generates controlled stimuli, and JudgeGPT
152 collects human judgements that are linked back to the generation parameters. Results from
153 this pipeline have been reported in peer-reviewed venues (Loth, Kappes, et al., 2026a, 2026d).

154 We recognise that a stimulus generator for misinformation research carries dual-use risk, and
155 we address it through the design rather than through disclaimers alone. The framework
156 generates short fragments for study delivery, not complete or distributable articles. It requires
157 the researcher to supply their own API credentials, so generation remains attributable to
158 an accountable party under the terms of the upstream providers, whose own usage policies
159 continue to apply. Human data collection through JudgeGPT is subject to institutional ethics
160 review. The corpus is archived on Zenodo under restricted access for academic use rather than
161 released openly (Loth, Kappes, et al., 2026c). The software is licensed under GPL-3.0, so
162 derivative work remains open to inspection.

163 The counterfactual matters here: the generative capability RogueGPT orchestrates is already
164 available to anyone through the same commercial APIs it calls, and it adds none. What does
165 not currently exist in the open is a reproducible, fully instrumented way for researchers to study
166 that capability under controlled conditions. Withholding the measurement infrastructure would
167 not reduce the availability of synthetic misinformation; it would leave the research community
168 without a shared basis for studying it.

169 AI Usage Disclosure

170 The software generates its research stimuli by calling commercial large language model APIs;
171 this is the documented function of the framework and is described throughout this paper.

172 For the development of the software and the preparation of this manuscript, AI-based coding
173 assistants were used for routine tasks such as code completion, refactoring suggestions, and
174 drafting of docstrings. All such output was reviewed, tested, and revised by the authors, who

175 take full responsibility for the content of the software and of this paper. The study design, the
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181 Origin Lens (Loth, Conceicao Rosario, et al., 2026), are available as open-source software.
182 The RogueGPT corpus is archived on Zenodo (Loth, Kappes, et al., 2026c). We thank the
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